GRAINS FUTURES STRATEGY — COMPLETE TECHNICAL DOCUMENT

1. Strategy Overview

Cross-sectional long/short grain futures strategy. Predicts relative performance between CORN, SOYBEAN, WHEAT_SRW, WHEAT_HRW.

2. Feature Engineering (Raw + Transformations)

Z-score normalization

```
z_t = (x_t - mean(x_{t-w:t})) / std(x_{t-w:t})
```

Non-linear transformation (tanh)

```
signal = tanh(z / 2)
```

Price features

Feature	Formula
mom_60	pct_change(price,60)
rev_5	-pct_change(price,5)
vol_20	std(returns,20)*sqrt(252)

Curve features

Feature	Formula
spread	C1 - C2
ratio	C1 / C2
change_20	spread_t - spread_t-20

Physical & COT

All physical variables are lagged and forward-filled.

```
inventory_change = inv_t - inv_t-1
receipts_change = rec_t - rec_t-1
cot_mm = mm / OI
```

```
cot_pm = pm / OI
```

3. Model

```
Core model:
X_core = [price + curve + COT]
y = 5d forward PnL
beta = Ridge(alpha=25)

Physical overlay:
X_phys = [inventory + receipts + cargill]
beta_phys = Ridge(alpha=1000)

prediction = X_core*beta + X_phys*beta_phys
```

4. Portfolio Construction

```
# Step 1: risk adjust
risk_adj_pred = pred / vol_20

# Step 2: cross-sectional demean
signal_i = risk_adj_pred_i - mean(risk_adj_pred)

# Step 3: normalize weights
weights = signal / sum(abs(signal))
```

Edge filter

```
edge = std(risk_adj_pred)
if edge < median(edge_history):
    weights = 0
```

5. Position Sizing & Volatility

```
returns = price.diff()

# realized vol
vol = std(returns,20)*sqrt(252)

# scaling
vol_scale = target_vol / vol

position = weights * vol_scale
```

6. PnL Calculation

```
# lag positions
gross_pnl_t = position_{t-1} * return_t * 5000
turnover = abs(position_t - position_{t-1})
```

```
cost = turnover * cost_bps / 10000

net_pnl = gross_pnl - cost

equity = cumprod(1 + net_pnl)
```

7. Instruments

Outright futures: main driver of PnL.

Inter-commodity spreads: overlay.

Calendar spreads: weak.

Crush spread: not tradable (missing legs).

8. Regime Behavior

Regime	Performance
Shock (drought/war)	Strong
Low vol	Weak
Trade war	Negative

9. Final Recommendation

Use static edge-filtered Ridge as candidate.

Use annual walk-forward Ridge as robustness check.